

SPATIALBLOCK: ENHANCING SPATIAL INTELLIGENCE IN LVLMS VIA SYNTHETIC BLOCK-STACKING PROBLEM

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ABSTRACT

Large Vision-Language Models (LVLMS) have achieved strong performance on diverse visual tasks, yet their ability to reconstruct and reason about the 3D structure of the scene depicted in 2D images – referred to as *spatial intelligence* – remains limited. Existing approaches attempt to address this gap by using real-scene spatial question answering datasets that require dense geometric annotations. However, constructing such labels is costly, time-consuming, and often noisy due to reliance on external perception modules. In this work, we propose a novel paradigm inspired by human cognitive development: learning foundational spatial skills through structured block-manipulation tasks. We introduce **SpatialBlock-15k**, a synthetic dataset of 15,000 block-stacking problems covering 3D-to-2D projection, viewpoint transformation, and structural combination. The dataset further incorporates controlled color modulation as visual cues to encourage anchor-based reasoning in visually complex conditions. Experiments demonstrate that LVLMS trained on our dataset through either direct answering or reasoning-based prediction significantly outperform baselines and generalize to real-world spatial tasks, despite the dataset’s synthetic and compact nature. Code and data are available at <https://github.com/rsoohyun/SpatialBlock>.

1 INTRODUCTION

Large Vision-Language Models (LVLMS) (Wang et al., 2024; Bai et al., 2025a; Zhu et al., 2025) have demonstrated remarkable performance across a wide range of 2D image understanding and reasoning tasks. Despite this progress, their ability to mentally reconstruct the 3D structure of a scene from 2D images – namely, *spatial intelligence* – remains limited. This shortcoming poses a fundamental bottleneck for deploying LVLMS in real-world applications that require robust spatial reasoning, such as autonomous driving (Tian et al., 2025) and robotics (Feng et al., 2025).

To address this limitation, prior work has extended model architectures by adding a specialized module for learning 3D features (Cheng et al., 2024; Wu et al., 2025) or designed tasks that directly require spatial reasoning (Yang et al., 2025; Yin et al., 2025) to encourage models to learn such capabilities. However, these approaches typically rely on real-world scene question answering datasets that require dense geometric and object-level annotations of the visual context. Constructing such dense labels is not only time-consuming and costly, but also noisy due to the frequent reliance on external modules (*e.g.*, segmentation or depth estimation models), thereby limiting their reliability and scalability.

In contrast to methods that depend on annotation-heavy real-scene data, we pursue a more efficient alternative inspired by the way humans develop spatial intelligence. Developmental

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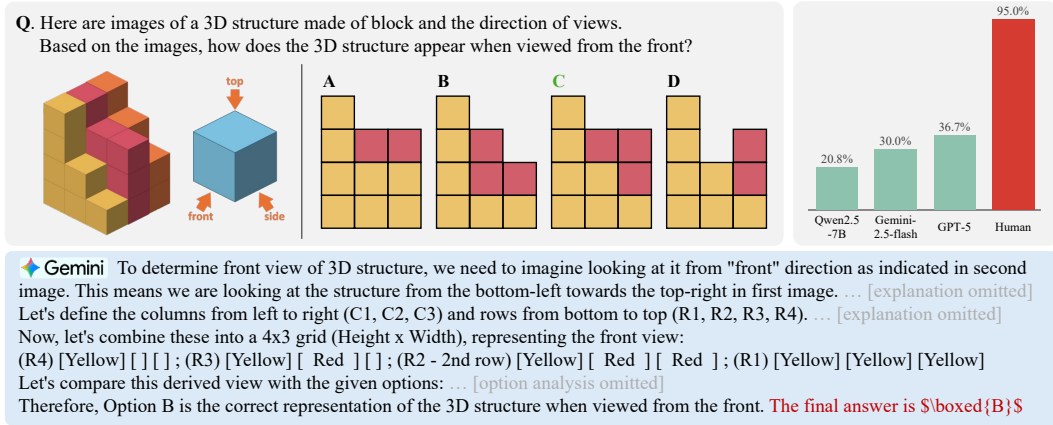


Figure 1: **Performance Gap Between Humans and LVLMs in Block-Stacking Problem.** The top row shows a block-stacking task that requires predicting a 2D projection from a given 3D structure. While humans achieve near-perfect accuracy, state-of-the-art LVLMs struggle even with basic spatial reasoning problems. Leading proprietary models exhibit flawed spatial logic, often producing incorrect interpretations of the structure.

psychology (Levine et al., 2012; Jirout & Newcombe, 2015) suggests that foundational spatial abilities are often cultivated through structured manipulative tools. Specifically, block play has been widely shown to enhance essential components of spatial cognition, including spatial integration and mental simulation (Caldera et al., 1999; Casey et al., 2008). Despite its simplicity, state-of-the-art LVLMs struggle with such problems, suggesting that they lack even fundamental levels of spatial intelligence (Fig. 1).

Motivated by this observation, we propose **SpatialBlock-15k**, a novel synthetic dataset of block-stacking problems designed to improve spatial intelligence. The dataset comprises three categories of spatial reasoning tasks: (1) 3D-to-2D projection, (2) viewpoint transformation, and (3) structural combination. To better approximate real-world spatial reasoning scenarios, where humans interpret scenes by anchoring on a specific object and reasoning relationships relative to it, we introduce color modulation as a visual cue, encouraging models to identify task-relevant elements in visually complex scenes. Unlike costly real-scene datasets, our dataset can be generated synthetically, enabling clean, scalable, and cost-effective data construction.

Experimental results show that LVLMs trained on SpatialBlock-15k, with either direct answer prediction or reasoning-based prediction, significantly outperform baseline models, even though the training data are entirely synthetic and relatively small in scale. These findings suggest that targeted training on fundamental spatial reasoning tasks can effectively enhance the spatial intelligence of LVLMs and provide a scalable alternative to annotation-heavy real-scene supervision.

In summary, our contribution is three-fold:

- We introduce a new perspective on enhancing spatial intelligence in LVLMs by shifting from annotation-heavy real-scene supervision to foundational spatial skill learning through structured synthetic tasks.
- We present **SpatialBlock-15k**, a scalable synthetic dataset of block-stacking problems spanning 3D-to-2D projection, viewpoint transformation, and structural combination, enhanced with controlled color cues for task-relevant reasoning.
- We show that training LVLMs on SpatialBlock-15k significantly improves spatial reasoning performance and generalizes effectively to real-world scenes, despite relying solely on synthetic data.

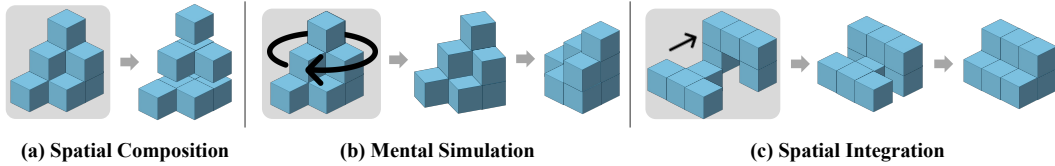


Figure 2: **Spatial Abilities Underlying Block-Stacking Reasoning.** Solving block-stacking tasks involves (a) inferring latent 3D composition, (b) reasoning under transformation, and (c) integrating multiple components into a coherent structure.

2 RELATED WORK

2.1 VISUAL SPATIAL REASONING IN LVLMS

Recent efforts to improve spatial intelligence in LVLMS focus on building a dedicated encoder or proposing a specialized training strategy. On the architectural side, some approaches modify the model by introducing spatial tokens into the vision encoder (Tong et al., 2024; Lou et al., 2025), incorporating depth-aware plugin modules (Cheng et al., 2024), or adding spatial encoders to learn 3D features (Wu et al., 2025). Other research focuses on the training process, designing hierarchical schemes that transition from spatial perception to complex reasoning (Ma et al., 2025a; Li et al., 2026; Liu et al., 2026) or encouraging the model to output cognitive maps that encode object position and orientation (Yang et al., 2025; Yin et al., 2025). Despite their effectiveness, these approaches generally require real-scene annotations or pseudo-3D signals from external modules, which limits their scalability and robustness.

2.2 TRAINING DATASETS FOR SPATIAL INTELLIGENCE

Various training datasets have been proposed to enhance the spatial intelligence of LVLMS, ranging from simple spatial relations to complex multi-step reasoning. Early benchmarks (Chen et al., 2024; Ma et al., 2025b) leverage external models (*e.g.*, object detection, segmentation, depth, or pose estimation) to extract 3D information and generate low-level QA pairs such as distance, orientation, and spatial relations. To foster higher-level logic, recent works (Yang et al., 2025; Ouyang et al., 2025) use 3D annotated video data to construct relative relation and direction reasoning, or more sophisticated tasks such as route planning or spatio-temporal appearance ordering through human annotation. MindCube (Yin et al., 2025) focuses on spatial mental modeling, which requires synthesizing a full scene from partial views or inferring arrangements under perspective shifts. Despite these advancements, a significant bottleneck remains: these approaches rely on dense, real-scene 3D annotations, which are labor-intensive and inherently noisy, necessitating costly human annotation.

3 SPATIALBLOCK-15K

We introduce **SpatialBlock-15k**, a scalable synthetic dataset designed to systematically probe and enhance foundational spatial abilities, moving away from the conventional annotation-intensive paradigm that relies on dense real-scene 3D labels. Specifically, we utilize block-stacking problems, which play a pivotal role in the early stages of human spatial cognitive development (Sec. 3.1). Furthermore, we incorporate controlled color variations as visual cues to better reflect real-world spatial reasoning conditions (Sec. 3.2).

3.1 BLOCK-STACKING PROBLEMS FOR SPATIAL INTELLIGENCE

Activities involving block-stacking require structural spatial understanding that extends beyond superficial visual pattern recognition. As illustrated in Fig. 2, solving such problems requires (a) spatial composition, the ability to reconstruct the complete 3D structure by deconstructing the visible configuration and inferring occluded blocks that must exist to

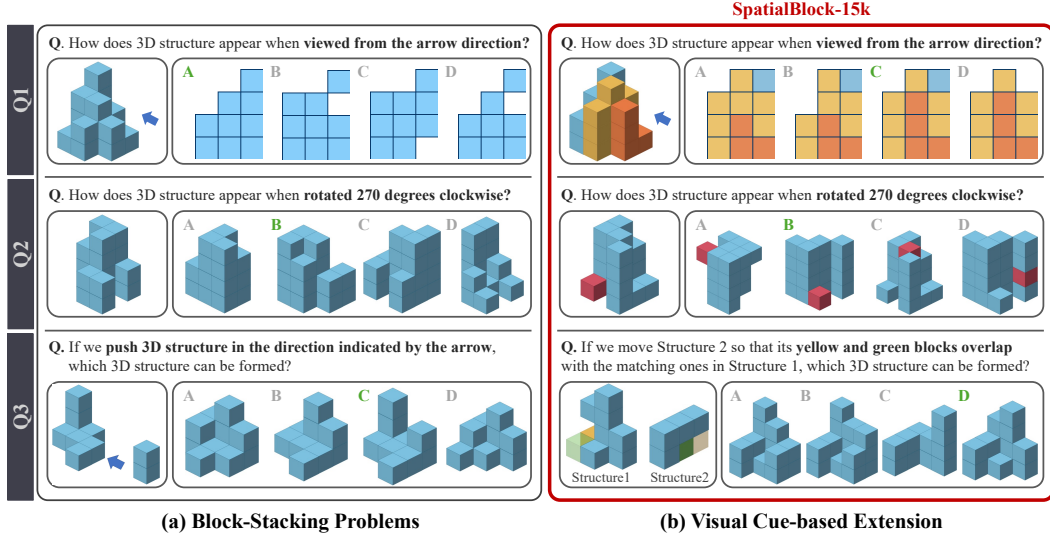


Figure 3: **SpatialBlock-15k**. (a) Three block-stacking tasks: 3D-to-2D projection (Q1), viewpoint transformation (Q2), and structural combination (Q3). (b) Visual cue-based extension with controlled color variations, where colors encode depth ordering (Q1), preserved structural roles across transformations (Q2), and correspondence cues for integration (Q3).

support it; (b) mental simulation, the capacity to simulate the outcome of transformations applied to a 3D structure; and (c) spatial integration, the ability to anticipate the resulting structure when multiple components are combined. Together, these abilities reflect core dimensions of spatial intelligence.

Building on these three abilities, we define three question types, each targeting a distinct aspect of spatial reasoning (Fig. 3(a)):

- Q1: 3D-to-2D projection** This task requires predicting the 2D appearance of a 3D structure from a specific viewing direction. Solving it demands the ability to internally reconstruct the 3D configuration and mentally transform it to the target viewpoint. The model must additionally perform depth-aware occlusion reasoning to determine which blocks remain visible in the final 2D projection.
- Q2: Viewpoint transformation** The problem inquires how a given 3D structure appears under self-rotation or viewpoint changes. The model should preserve structural consistency across transformations, maintaining correct relative positions of all components without distortion, disappearance, or unintended displacement.
- Q3: Structural combination** This question asks how the resultant integration is formed when two 3D structures are combined. It requires simulating how individual components interact, identifying contact interfaces, and subsequently inferring the coherent global structure that emerges from their integration.

Collectively, these question types form the foundation of our SpatialBlock-15k dataset, providing a structured framework for evaluating and training spatial reasoning in LVLMs.

3.2 VISUAL CUE-BASED EXTENSION

In the previous section, we constructed tasks that require understanding 3D structures composed of single-color blocks. Since color does not provide a discriminative feature, these tasks demand that the model focus on purely geometric relationships, such as the relative positions of blocks and appearance changes under viewpoint transformations.

However, real-world scenes typically consist of complex environments with diverse objects, rather than visually uniform ones. Given this intricacy, humans tend to analyze the global

scene configurations by selecting a specific object as an anchor and interpreting relationships relative to it (Land & Hayhoe, 2001). That is, visually salient cues serve as the reference points to infer changes and relations.

Motivated by this cognitive characteristic, we extend the monochromatic block-based tasks introduced in Sec. 3.1 by incorporating color as an additional cue. Here, color is not used merely for visual diversity but as a functional guidance that reveals depth information, structural correspondences, and contact regions. This design encourages the model to reconstruct the global structure and track transformations by using specific blocks as reference points. Specifically, we extend the three question types as follows (Fig. 3(b)):

Q1: 3D-to-2D projection with depth cues To facilitate depth-aware reasoning, different colors are assigned based on depth from a given viewpoint. Thus, the task requires not only matching the silhouette but also tracking front-back relationships using color information. Color functions as an explicit indicator of depth ordering and is designed to clearly reflect the hierarchical organization of the 3D structure.

Q2: Viewpoint transformation with anchor blocks Among the blocks composing the 3D structure, one anchor block is assigned a distinct color. This block must maintain the same structural role before and after transformation. Accordingly, the task requires not only recognizing geometric structure but also understanding anchor-relative relationships that are preserved through transformation.

Q3: Structural combination with overlapping blocks The attachment location between two structures is represented not by arrows but by the overlap of colored blocks indicated by transparent regions. Each structure is presented as a separate image, so aligning the positions of same-colored blocks across images is needed to predict the combined structure. Color thus acts as a shared reference point linking two independent inputs, inducing reasoning inter-structural integration.

This visual cue-based extension is designed to maintain the requirement for understanding the 3D structure itself while additionally enabling the learning of scene organization and relational reasoning centered around anchor objects, as required in realistic visual contexts. Consequently, it leads to meaningful improvement in real-scene spatial reasoning performance, as empirically demonstrated in Sec. 5.3.2. Finally, the SpatialBlock-15k dataset comprises three question types with visual cue-based extensions, each containing 5k samples. Further details on data construction can be found in Sec. A.

4 METHOD

To foster the spatial intelligence of LVLMs using SpatialBlock-15k, we propose two training strategies, targeting different aspects of the model’s capabilities: (1) direct answer prediction, which focuses on cultivating rapid inference by mapping visual inputs directly to their corresponding answers (Sec. 4.1), and (2) reasoning-based prediction, which encourages the model to explain its intermediate logical paths that lead to a final answer (Sec. 4.2).

4.1 DIRECT ANSWER PREDICTION MODEL

To maximize the model’s capacity for instantaneous spatial problem-solving, we train it with supervision only on the ground-truth answer sequence $y = \{y_1, y_2, \dots, y_T\}$. Specifically, given a textual query q and image v , we optimize model parameters θ to predict the next token y_i , given the preceding context $y_{(1:i-1)}$, by minimizing standard cross-entropy loss:

$$\mathcal{L}_{\text{ce}}(\theta) = - \sum_i \log P(y_i | y_{(1:i-1)}, q, v). \quad (1)$$

This strategy directly optimizes answer precision, thereby establishing a foundation for high-fidelity spatial problem-solving proficiency.

4.2 REASONING-BASED PREDICTION MODEL

To go beyond simple answer prediction, we adopt a second strategy that enables the model to articulate its internal logic. To ensure a solid foundation for spatial intelligence, we first initialize the model through supervised fine-tuning to establish basic task-solving ability, then employ reinforcement learning to facilitate high-order structural reasoning.

For model initialization, we train the model with Low-Rank Adaptation (LoRA) instead of full-parameter fine-tuning, as full fine-tuning often degrades the inherent reasoning ability. This lightweight adaptation maintains the model’s Chain-of-Thought (CoT) reasoning capabilities while aligning it with our dataset. Notably, we deviate from the conventional “cold-start” phase that relies on synthesized trajectories from larger teacher models (Guo et al., 2025; Li et al., 2026). The rationale behind this decision is that even state-of-the-art proprietary models exhibit relatively low accuracy on our dataset (Tab. 1), making it infeasible to construct reliable high-quality trajectories. Our empirical analysis (Sec. 5.3.4) demonstrates a clear performance advantage in preserving the model’s reasoning ability through LoRA-based tuning. This approach provides a more effective initialization for RL training than attempting to recover the thinking ability via a low-quality cold-start phase.

After LoRA-based initialization, we directly optimize the model using reinforcement learning via Group Relative Policy Optimization (GRPO). We design a multi-objective reward function that evaluates response correctness and reasoning trace quality with three components:

$$\mathcal{R}(y, o) = \mathcal{R}_{\text{acc}}(y, o) + \mathcal{R}_{\text{format}}(o) + \mathcal{R}_{\text{len}}(o), \quad (2)$$

where o is a model-generated prediction given a question $c = (q, v)$ with ground-truth answer y . The accuracy reward $\mathcal{R}_{\text{acc}}$ verifies whether the final answer extracted from o matches y .

Next, the format reward $\mathcal{R}_{\text{format}}$ enforces the required CoT structure, with a comprehensive reasoning trace followed by the final answer enclosed within `<answer>` and `</answer>` tags. Lastly, to prevent overly brief or excessively verbose output, a length reward $\mathcal{R}_{\text{len}}$ requires the response length L to satisfy $50 < L < 1024$. For simplicity and stability, all components are implemented in binary rewards (0 or 1) based on whether each condition is satisfied.

Given this reward, for each question c , the old policy model $\pi_{\theta_{\text{old}}}$ samples a group of candidate responses $\{o_1, o_2, \dots, o_G\}$. Each response o_i receives a reward $r_i = \mathcal{R}(y, o_i)$, from which we compute the advantage $A_i = \frac{r_i - \text{mean}(r_1, r_2, \dots, r_G)}{\text{std}(r_1, r_2, \dots, r_G)}$. The model is then updated by maximizing the following objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{c, \{o_i\}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(\frac{\pi_{\theta}(o_i|c)}{\pi_{\theta_{\text{old}}}(o_i|c)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|c)}{\pi_{\theta_{\text{old}}}(o_i|c)}, 1 \pm \varepsilon \right) A_i \right) - \beta \text{KL}[\pi_{\theta} || \pi_{\text{ref}}] \right], \quad (3)$$

where ε and β are hyper-parameters, and $\text{KL}[\pi_{\theta} || \pi_{\text{ref}}]$ is the KL divergence between the policy model π_{θ} and reference model π_{ref} .

5 EXPERIMENTS

We conduct multiple experiments to validate the effectiveness of SpatialBlock-15k in fostering spatial intelligence. We first describe the experimental setup, including training details and evaluation protocols (Sec. 5.1). We then compare our models with prior methods across several spatial reasoning benchmarks (Sec. 5.2). Next, we perform ablation studies to assess the effects of each component in dataset construction and training (Sec. 5.3). Lastly, we provide an analysis of the generalizability of our proposed dataset (Sec. 5.4).

5.1 EXPERIMENTAL SETTINGS

We adopt Qwen2.5-VL-3B (Bai et al., 2025b), Qwen2.5-VL-7B (Bai et al., 2025b), Qwen3-VL-4B (Bai et al., 2025a), and InternVL3-2B (Zhu et al., 2025) as baseline models for our framework. To distinguish the training strategy, we append a suffix to the model name: *SpatialBlock-direct* denotes the model that outputs the answer directly (Sec. 4.1), while *SpatialBlock-reason* refers to the model that performs reasoning before producing the final answer (Sec. 4.2). Both variants are trained using the entire SpatialBlock-15k dataset.

Table 1: **Main Results.** For each dataset, **bold** and underlined indicate the best and second-best performance, respectively. * represents in-domain results where the model was trained on the benchmark. We re-evaluated all previous works for fair comparison.

Model	# Data	In-domain	Out-of-domain				
		SB-Bench	MindCube	MMSI-Bench	SPBench	MMMU	Overall
Proprietary Models							
GPT-5 (Singh et al., 2025)	-	43.8	56.7	42.8	43.0	77.6	55.0
Gemini-2.5-flash (Comanici et al., 2025)	-	36.4	51.0	32.2	44.3	58.8	46.6
Claude-Sonnet-4.5 (Anthropic, 2025)	-	46.3	34.0	32.7	43.9	69.2	43.9
Open-Source Models							
LLaVA-OneVision-7B (Li et al., 2024)	-	23.8	39.0	28.4	43.4	42.6	39.2
InternVL3-8B (Zhu et al., 2025)	-	22.7	42.7	25.1	46.1	48.0	40.5
Llama-3.2-11B-Vision-Instruct (Dubey et al., 2024)	-	23.0	29.5	24.5	28.2	40.7	30.7
Spatial Specialist Models							
Qwen2.5-VL-3B Based Models							
Qwen2.5-VL-3B-Instruct (Bai et al., 2025b)	-	29.9	39.1	26.1	40.0	45.6	37.7
SpaceQwen (Chen et al., 2024)	2B	30.6	38.8	24.5	40.8	45.6	37.4
SpatialMLLM (Wu et al., 2025)	120K	30.0	36.3	25.4	40.8	40.4	35.7
SpatialLadder-3B (Li et al., 2026)	25K	25.5	46.4	26.0	76.6*	47.3	39.9
Spatial-SSRL-3B (Liu et al., 2026)	81K	29.5	41.0	24.4	19.5	45.2	32.5
SpatialBlock-3B-direct (Ours)	15K	94.8	49.1	28.1	40.4	46.4	41.0
SpatialBlock-3B-reason (Ours)	15K	90.2	45.3	29.2	43.7	46.7	41.2
Qwen2.5-VL-7B Based Models							
Qwen2.5-VL-7B-Instruct (Bai et al., 2025b)	-	20.8	31.3	27.4	46.7	46.6	38.0
SpaceR (Ouyang et al., 2025)	150K	26.8	31.6	27.4	53.1	55.4	41.9
Spatial-SSRL-7B (Liu et al., 2026)	81K	27.7	35.1	27.5	23.4	48.9	33.7
SpatialBlock-7B-direct (Ours)	15K	95.0	48.8	28.4	48.7	45.2	42.8
SpatialBlock-7B-reason (Ours)	15K	77.5	37.9	29.7	50.9	55.5	43.5
Qwen3-VL-4B Based Models							
Qwen3-VL-4B-Instruct (Bai et al., 2025a)	-	17.0	26.2	27.1	44.5	47.5	36.3
Spatial-SSRL-4B (Liu et al., 2026)	81K	34.8	34.6	28.3	23.1	50.5	34.1
SpatialBlock-4B-direct (Ours)	15K	95.7	51.3	27.8	45.9	47.2	43.1
SpatialBlock-4B-reason (Ours)	15K	94.5	41.1	28.3	43.7	49.6	40.7
InternVL3-2B Based Models							
InternVL3-2B-Instruct (Zhu et al., 2025)	-	19.0	32.1	27.0	33.0	41.3	33.4
SpatialBlock-2B-direct (Ours)	15K	97.2	51.4	27.3	41.3	42.4	40.6
SpatialBlock-2B-reason (Ours)	15K	78.3	38.9	27.8	45.6	47.3	39.9

To evaluate the spatial reasoning capability, we conduct experiments on five benchmarks spanning both in-domain and out-of-domain settings. For in-domain evaluation, we introduce SB-Bench, the held-out test split of SpatialBlock-15k, comprising 600 questions. For out-of-domain evaluation, we assess generalization to real-world scenarios on four benchmarks: MindCube (Yin et al., 2025), MMSI-Bench (Yang et al., 2026), and SPBench (Li et al., 2026) for spatial reasoning, and MMMU (Yue et al., 2024) for general visual perception. Across all benchmarks, we restrict evaluation to multiple-choice questions to align with our training answer format. Additional numerical results are provided in Sec. C.1.

5.2 MAIN RESULTS

Tab. 1 shows our method’s performance on spatial reasoning benchmarks. Notably, despite being trained with only 15k synthetic SpatialBlock samples, our model consistently outperforms existing spatial specialists on real-scene benchmarks. This result highlights that our block-stacking formulation provides an effective training signal for spatial reasoning.

For **SpatialBlock-direct** models, we observe substantial gains, particularly on MindCube, which requires mental simulation. Among Qwen-based models, the 3B model outperforms the state-of-the-art SpatialLadder by 2.7%, while the 7B model shows an even larger gain of 17.6% over its backbone. The 4B model further improves upon its backbone by 25.1%, achieving 51.3%, the best open-source performance. Beyond Qwen, our method also improves InternVL3, demonstrating its generalizability across model scales and families.

The reasoning-enhanced variant, **SpatialBlock-reason**, further demonstrates strong results on MMSI-Bench, which requires complex logical inference. The 3B model outperforms SpatialLadder-3B by 3.2% using a simpler training pipeline based solely on synthetic block-stacking data, whereas the 7B model outperforms SpaceR and Spatial-SSRL with only 15K training samples. Beyond final-answer accuracy, our model also improves reasoning quality, achieving a higher alignment score of 21.1 with ground-truth reasoning steps, compared to 17.8 for the baseline. (See Sec. B.4 for details.) On SPBench, our models consistently

Table 2: **Ablation Studies.** * denotes in-domain results from benchmark-trained models.

Model	# Data	In-domain	Out-of-domain				
		SB-Bench	MindCube	MMSI-Bench	SPBench	MMMU	Overall
Qwen2.5-VL-3B-Instruct (Bai et al., 2025b)	-	29.9	39.1	26.1	40.0	45.6	37.7
SpatialBlock-3B-direct (Ours)	15K	94.8	49.1	28.1	40.4	46.4	41.0
SpatialBlock-3B-reason (Ours)	15K	90.2	45.3	29.2	43.7	46.7	41.2
(a) Ablation on question types							
SpatialBlock-3B-reason (Q1 only)	15K	48.0	41.8	27.6	42.8	43.2	38.8
SpatialBlock-3B-reason (Q2 only)	15K	51.5	38.8	24.2	42.1	47.1	38.0
SpatialBlock-3B-reason (Q3 only)	15K	40.8	41.3	25.3	43.2	46.5	39.1
(b) Ablation on visual cues							
SpatialBlock-3B-direct (w/o visual cues)	15K	61.7	41.2	26.2	38.4	46.6	38.1
SpatialBlock-3B-reason (w/o visual cues)	15K	55.8	38.0	24.4	44.8	48.8	39.0
(c) Ablation on task design							
SpatialBlock-3B-direct (SpatialLadder-26k)	15K	24.7	44.5	27.2	72.1*	45.3	47.3
SpatialBlock-3B-reason (SpatialLadder-26k)	15K	28.0	44.9	27.5	68.8*	44.7	46.5
(d) Ablation on model initialization							
SpatialBlock-3B-reason (No Init.)	15K	67.8	41.3	25.4	43.5	46.7	39.2
SpatialBlock-3B-reason (Full-SFT Init., Qwen2.5-VL-7B)	15K	96.5	43.0	27.6	47.8	44.7	40.8
SpatialBlock-3B-reason (Full-SFT Init., GPT-5)	15K	75.3	41.7	28.0	44.3	48.1	40.5

improve relative spatial reasoning despite such questions not being explicitly included in the training data, demonstrating strong generalization beyond the supervised task distribution.

In summary, direct models excel on benchmarks that require canonical 90-degree view-point transformations, while reasoning models perform better on tasks requiring diverse viewpoint changes and multi-image reasoning. Also, despite being trained without any real-scene images, our models maintain stable performance on the general visual understanding benchmark MMMU, indicating that SpatialBlock training enhances 3D spatial reasoning without degrading broader visual perception.

5.3 ABLATION STUDIES

5.3.1 ABLATION ON QUESTION TYPES

We evaluate models trained on a single question type (Tab. 2(a)). Each question type targets a distinct aspect of spatial reasoning: Q1 focuses on viewpoint changes, Q2 emphasizes spatial transformations such as 90-degree rotations, and Q3 requires integrating spatial information across multiple perspectives. The single-type variants exhibit different performance patterns across benchmarks, suggesting that each type provides distinct spatial supervision. However, while single-type variants show limited gains, training on all three types achieves the best performance across most benchmarks. These results indicate that the three types provide complementary supervision, and combining them leads to more robust spatial reasoning and generalization.

5.3.2 ABLATION ON VISUAL CUES

We introduce color manipulation as a visual cue to approximate how humans interpret real-world scenes using visually salient reference objects (Sec. 3.2). To evaluate its effectiveness, we compare models trained with and without visual cues. As shown in Tab. 2(b), removing visual cues leads to performance drops in *direct* and *reason* models of 7.9% and 7.3% in MindCube, and 1.9% and 4.8% in MMSI-Bench, respectively. These results suggest that color cues help models better understand complex scenes and mimic human scene perception, improving spatial reasoning in real-world environments.

5.3.3 ABLATION ON TASK DESIGN

In order to analyze the effect of the task design itself more thoroughly, we train a model on SpatialLadder-26k using the identical training recipe. For a fair comparison, we use 11.5k multiple-choice questions matching our response format and train for 1.3 epochs to match the same training size. As shown in Tab. 2(c), this model underperforms ours on most benchmarks except the in-domain benchmark SPBench, suggesting that our performance gains mainly stem from the proposed block-stacking task design.

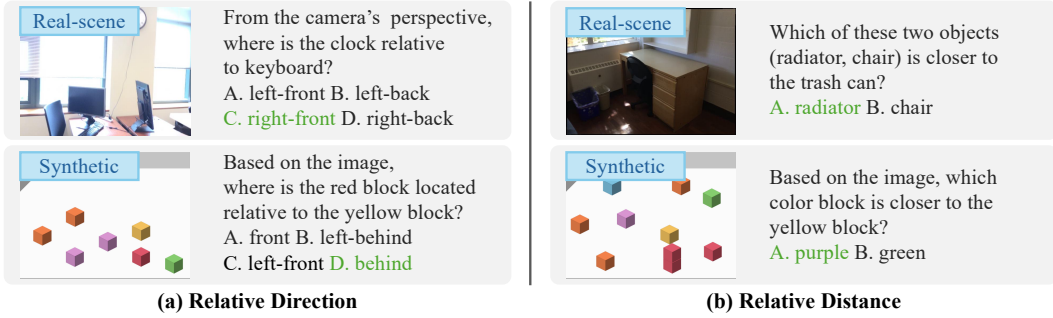


Figure 4: **Synthetic-Real Dataset**. Widely used spatial question types, (a) relative direction and (b) relative distance, are adapted into a block-based environment while preserving their semantic structure.

Table 3: **Comparison of Training on SpatialBlock and Synthetic-Real**. While Synthetic-Real improves in-domain performance, training on SpatialBlock leads to better generalization on out-of-domain benchmark datasets. All experiments are conducted on Qwen2.5-VL-3B (Bai et al., 2025b), and in-domain performance is evaluated on SPBench.

Train Dataset	In-domain		Out-of-domain		
	Relative Direction	Relative Distance	MindCube	MMSI-Bench	MMM
Baseline	31.4	74.1	39.1	26.1	45.6
Synthetic-Real	43.4	76.9	32.8	25.5	43.6
SpatialBlock	31.4	67.4	49.1	28.1	46.4

5.3.4 ABLATION ON MODEL INITIALIZATION FOR REASONING MODEL

To train a reasoning-capable model via reinforcement learning, we adopt a LoRA-based initialization strategy to enhance baseline spatial intelligence while preserving inherent reasoning ability. Tab. 2(d) compares different initialization strategies and their effects on performance. Without prior fine-tuning, performance drops noticeably across benchmarks, indicating the importance of establishing basic spatial understanding. We further compare our approach with commonly used cold-start phases based on full-parameter tuning (Guo et al., 2025; Li et al., 2026). While cold-start training can improve instruction following and achieve higher accuracy convergence, generating reliable trajectories for our tasks is challenging: both a larger model (Qwen2.5-VL-7B) and a proprietary model (GPT-5) frequently produce inaccurate block descriptions and reasoning. Consequently, training with these noisy reasoning traces yields only marginal gains. In contrast, LoRA-based initialization provides substantial improvements, offering a strong starting point for reinforcement learning with both basic spatial intelligence and reliable instruction-following ability.

5.4 GENERALIZABILITY OF BLOCK-STACKING QUESTIONS

As shown in Tab. 1, our models trained on SpatialBlock-15k generalize well to real-scene benchmark datasets despite involving question types that differ from those commonly used. This observation raises the question of whether our block-stacking tasks provide a more effective learning signal than conventional spatial question formulations. To investigate this, we construct *Synthetic-Real*, a dataset of the same size as SpatialBlock-15k, which adapts two widely used spatial question types – relative direction and distance – into our block-based environment (Fig. 4). Although both datasets can be generated with comparable effort, the results in Tab. 3 reveal that the model trained on Synthetic-Real achieves higher accuracy only on in-domain tasks and performs poorly on out-of-domain benchmark datasets. In contrast, our model consistently improves across benchmarks. These results suggest that SpatialBlock tasks promote a more holistic understanding of 3D scenes, leading to more generalizable and transferable spatial reasoning than standard spatial supervision.

6 CONCLUSION

In this paper, we revisited the problem of improving spatial intelligence in large vision-language models from a new perspective: learning foundational spatial skills through structured tasks rather than relying on annotation-heavy real-scene supervision. To this end, we introduced **SpatialBlock-15k**, a novel synthetic dataset of block-stacking problems, including 3D-to-2D projection, viewpoint transformation, and structural integration. The dataset further incorporated controlled color variation as visual cues to encourage anchor-based reasoning, mimicking how humans interpret scenes by using specific objects as reference points. Using this dataset, we developed two specialized models: **SpatialBlock-direct** for immediate answer generation and **SpatialBlock-reason** for detailed logical inference. Experimental results demonstrated that our models consistently outperform existing spatial specialists on various real-world spatial benchmarks. Furthermore, extensive analyses show that the learned capabilities generalize across tasks and domains, indicating that the models capture the underlying spatial structure of scenes. These findings highlighted that robust spatial intelligence can be achieved through training on carefully designed synthetic spatial tasks, offering a scalable and effective path for improving spatial reasoning in LVLMs.

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A DETAILS ON CONSTRUCTION AND EVALUATION OF SPATIALBLOCK-15K

A.1 DATA CONSTRUCTION PROCESS

All 3D structures in SpatialBlock-15k are constructed on a 3×3 spatial grid, where each grid position can contain between one and four vertically stacked blocks. The height of each stack defines the underlying 3D configuration. To avoid ambiguity when interpreting rendered structures, we restrict the set of valid configurations so that the number of blocks at every grid position can be reliably inferred from the rendered image. In particular, configurations that have severe occlusion – where blocks are completely hidden behind others from the viewing perspective – are excluded. This constraint ensures that the global structure remains visually interpretable while still allowing a diverse set of configurations.

Based on the resulting valid structures, we generate the three types of tasks illustrated in Fig. 3. For each task, the distractor choices are carefully constructed to be structurally plausible while incorrect, preventing simple heuristics and encouraging models to reason about the spatial relationships within the structure.

A.1.1 Q1: 3D-TO-2D PROJECTION WITH DEPTH CUES

In this task, the model must predict the correct 2D projection (e.g., front or side view) of a given 3D block structure. The wrong choices are generated through controlled perturbations of the correct projection:

- **Block-removal perturbation.** A visible top block from one column is removed, producing a projection that corresponds to a slightly reduced stack height while remaining structurally plausible.
- **Color-consistency perturbation.** For a column containing multiple colors, the color of one block is replaced with that of a neighboring block in the same column. This preserves the geometric structure but violates the correct color arrangement.
- **Combined perturbation.** A hybrid modification that simultaneously removes a block from one column and alters the color assignment in another column.

During choice generation, perturbations are only applied to columns that contain visible blocks and sufficient color diversity, ensuring that distractors remain meaningful and visually distinguishable.

A.1.2 Q2: VIEWPOINT TRANSFORMATION WITH ANCHOR BLOCKS

This task evaluates whether models can reason about global spatial transformations of the structure. Given an original structure, one transformation is randomly selected from the following set: 90° , 180° , or 270° rotations, and flipping vertically. A specific visible block in the original structure is highlighted, and the model needs to identify the corresponding block position after the transformation. To ensure valid supervision signals, only blocks that remain visible after transformation are selected as colored blocks.

The answer choices consist of four candidates:

- the correctly transformed structure with the correct mapped block position (correct answer),
- the original structure with a different block highlighted,
- the correctly transformed structure with an incorrectly highlighted block,
- or a structure generated through a different transformation.

A.1.3 Q3: STRUCTURAL COMBINATION WITH OVERLAPPING BLOCKS

This task focuses on reasoning about the union when two block structures are combined. Each problem instance consists of two structures and a specified attachment point indicated

by overlapping color. The goal is to determine the correct outcome of the combination. During dataset construction, we sample a possible combination from the given 3D structures and compute their first contact location as the overlap region.

The distractor choices are constructed by choosing a 3D configuration where one component is replaced, and the other remains unchanged, resulting in a different structure. By ensuring that each distractor differs only in one of the two input structures, the choices remain visually similar to the correct answer while requiring the model to reason about the precise spatial interaction between the specific pair of structures.

A.2 HUMAN EVALUATION ON SB-BENCH

For human evaluation, we randomly select 30 samples from SB-Bench and collect responses from 31 participants. As illustrated in Fig. 1, humans achieve 95% accuracy on average, demonstrating the task’s simplicity.

B TRAINING IMPLEMENTATION DETAILS

B.1 EARLY STOPPING FOR SPATIALBLOCK-DIRECT

To prevent overfitting to the block structure during direct model training, we selected the model from an intermediate iteration in which test accuracy exceeded 93%, specifically the iteration immediately before it began to decrease.

B.2 REASONING PROMPT USED FOR REINFORCEMENT LEARNING

The detailed instructions for reinforcement learning are shown in Fig. 5. The prompt guides the model to output reasoning steps in a numbered sequence and gives the final answer between `<answer>` and `</answer>` tags. We do not use the extra `<reason>` tag to reduce the load of learning the format.

RL prompt

Please think about this question step by step and provide your detailed reasoning in a numbered reasoning sequence (1., 2., 3.,...). Then give your final answer between the `<answer>` `</answer>` tags. Output must be EXACTLY in this format (including tags):

...

`<answer>X</answer>`

Please provide only the single option letter (e.g., A, B, C, D, etc.) within the `<answer>` `</answer>` tags.

Figure 5: **Instructions for Reinforcement Learning.** The model is instructed to provide a step-by-step thinking process, and then the final answer between answer tags.

B.3 HYPERPARAMETERS USED FOR TRAINING

We apply LoRA-based supervised fine-tuning for SpatialBlock-reason, using rank 8 for the 3B and 2B models and rank 16 for the 7B and 4B models. For reinforcement learning, we use EasyR1 (Yaowei Zheng et al., 2025) as our base code and follow its default hyperparameters. We set the batch size to 512 with 5 rollouts, while extending the maximum prompt length to 4096 to accommodate multi-image training. All training was conducted for 1 epoch using SpatialBlock-15k.

B.4 REASONING QUALITY EVALUATION PROTOCOL

To further assess reasoning quality, we utilize MMSI-Bench with its ground-truth rationales. GPT-5 scores each reasoning’s alignment with each ground-truth step as 1 (present), 0.5 (implicit), or 0 (incorrect), and the final score is averaged and scaled to 100. The detailed instructions are shown in Fig. 6.

GPT-eval prompt

TASK : Compare the reasoning of Model 1 and Model 2, given the Ground Truth (GT) reasoning.
 First, break the GT reasoning into its core logical step.
 For each GT reasoning steps, label each model's reasoning as present/implicit/incorrect.
 Finally output the final winning model between <answer> and </answer>. If they are both equally bad or good, output None.
 GT reasoning : {gt} Model 1 : {model_1} Model 2 : {model_2}

Format :

<GT> Step1: [reasoning], ... </GT> <Model 1> Step1 : [label], ... </Model 1> <Model 2> Step2 : [label], ... </Model 2>
 <answer> Model X </answer>

Figure 6: **Instructions for Reasoning Quality Evaluation.** We evaluate the quality of model-generated reasoning by measuring its alignment with each step of the corresponding ground-truth rationale.

C ADDITIONAL RESULTS

C.1 ADDITIONAL RESULTS ON NUMERICAL TASKS

Table 4: **Results on Numerical Tasks.** Results on tasks from SPBench.

Model	Absolute distance	Object size	Object counting	Average
Qwen2.5-VL-3B-Instruct	30.9	17.6	35.1	27.9
SpatialBlock-3B-direct	23.2	19.9	72.1	38.4
SpatialBlock-3B-reason	21.4	38.3	43.3	34.3

We evaluate our model on numerical prediction tasks from SPBench (Tab. 4). Despite being trained only on multiple-choice questions, our model improves in counting and size estimation, as our dataset enhances the model’s ability to individuate objects and infer their scales. In contrast, it is less effective at absolute distance because our method prioritizes structural understanding and spatial integration over precise metric regression.

C.2 ROBUSTNESS TO VIEWING ANGLE VARIATIONS

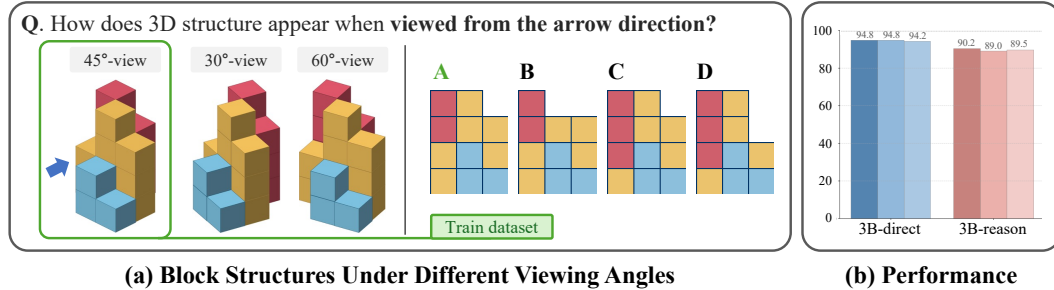


Figure 7: **Performance Across Viewing Angles.** (a) The same block structures are rendered from different viewing angles. (b) Performance of models across viewing angles; the bars correspond to results under 45°, 30°, and 60° views, respectively.

To further evaluate whether the model learns the underlying 3D structure rather than memorizing appearance patterns from a fixed viewpoint, we conduct a viewpoint generalization experiment. Our SpatialBlock-15k dataset is constructed using blocks rendered from a 45° viewing angle, which is fixed during training. We generate the same problems using identical block configurations rendered from different viewing angles (30° and 60°) (Fig. 7(a)). As shown in Fig. 7(b), both the direct answering and reasoning-based models maintain consistent performance across these viewpoints. This suggests that the models capture the underlying structure of the blocks, demonstrating genuine enhancement in spatial ability.

C.3 COLD-START DATA QUALITY

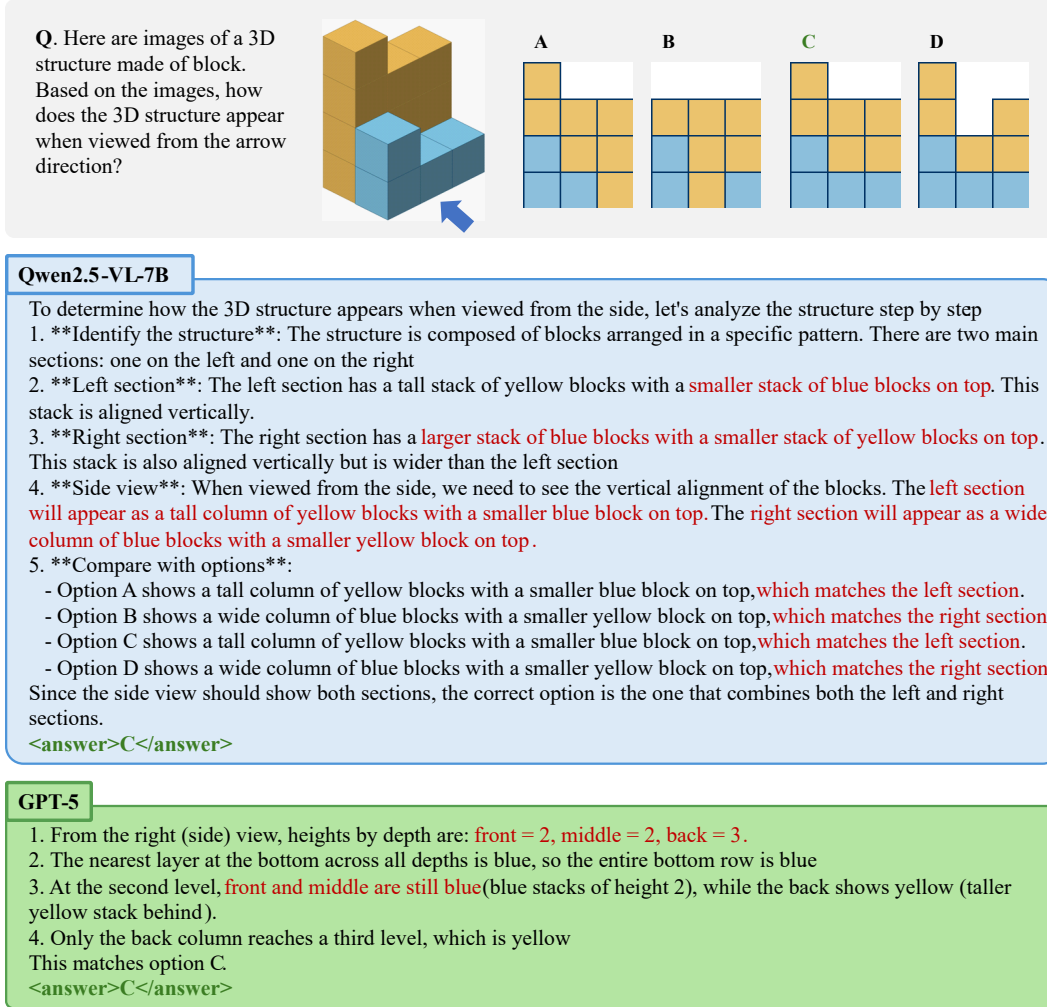


Figure 8: **Generated Cold-Start Data Sample.** The reasoning trace extracted from Qwen2.5-VL-7B includes incorrect information about the given 3D structure, even though the model answers correctly.

As mentioned in Sec. 5.3.4, full-parameter fine-tuning with a cold-start phase shows marginal improvement compared with LoRA-based initialization. We hypothesize that this is due to the low quality of the cold-start reasoning trace. Fig. 8 shows a reasoning example extracted from Qwen2.5-VL-7B. Although we choose the one with correct answers, the model fails to explain the 3D structure correctly, particularly miscounting the blocks. Furthermore, despite the attempt to evaluate all options, it generates the same explanation for distinct options, leading to a random final choice.

We also observed that even proprietary models such as GPT-5 frequently generate inaccurate block descriptions and reasoning (Fig. 8), highlighting the difficulty of producing stable trajectories even for top-tier models on our tasks. Using these GPT-5-generated labels for cold-start SFT still underperforms ours, especially on spatial reasoning benchmarks (Tab. 2(d)), which supports our claim that low-quality teacher reasoning can serve as noisy supervision.

C.4 QUALITATIVE RESULT

Q. Based on these three images showing the same scene from different viewpoints (back, left, and right): From the viewpoint presented in image 2, is there a yellow-plastic container behind the grey parking meter?

A. Yes B. No.



(a) Baseline model (Qwen2.5-VL-3B)

1. First, let's analyze the information provided in **image 2**: The image shows a grey parking meter on a sidewalk. There is a red sign with white text to the left of the parking meter.
 2. Next, let's consider the question: The question asks if there is a yellow-plastic container behind the grey parking meter. We need to determine if such a container exists based on the information given in **image 2**.
 3. Now, let's examine the details in **image 2**: The focus is on the grey parking meter and the red sign. There is no mention or indication of any yellow-plastic container in the image.
 4. Finally, let's draw a conclusion: Since there is no mention of a yellow-plastic container in **image 2**, we cannot confirm its presence. Therefore, the correct answer is that there is no yellow-plastic container behind the grey parking meter.

<answer>B</answer>

(b) SpatialBlock-3B-reason (without visual cue)

Let's analyze the images step by step to determine if there is a yellow-plastic container behind the grey parking meter from the viewpoint presented in image 2.

1. **Image 1**: This image shows two yellow portable toilets (porta potties) on a sidewalk. There is no mention of a yellow-plastic container in this image.
2. **Image 2**: This image shows a close-up of a grey parking meter on a sidewalk. There is a red and white sign in the background, but no yellow-plastic container is visible in this image.
3. **Image 3**: This image shows a close-up of a yellow plastic container, which appears to be a porta potty. The container is positioned on a sidewalk, and there is a tree in the background.

Based on the analysis: **Image 1 does not show a yellow-plastic container. Image 2 does not show a yellow-plastic container. Image 3 shows a yellow-plastic container, which is a porta potty.**

From the viewpoint presented in image 2, there is no yellow-plastic container visible. Therefore, the answer is:

<answer>B</answer>

(c) SpatialBlock-3B-reason

1. Observing **image 1**, we can see two yellow-plastic containers standing side by side on a sidewalk.
2. Looking at **image 2**, we see a grey parking meter on a sidewalk with a yellow sign in the background.
3. **Comparing image 2 to the scene in image 1, we can see that the yellow sign in image 2 matches the yellow container in image 1.**
4. Therefore, based on the matching yellow sign and container, we can conclude that there is a yellow-plastic container behind the grey parking meter in image 2.
5. The answer is A.

<answer>A</answer>

Figure 9: **Qualitative Comparison of Reasoning Traces.** (a) Baseline only describes a single view. (b) Training without visual cues attempts multi-image reasoning but fails to identify the shared object. (c) Our model uses the “yellow container” as an anchor to integrate views and reach the correct answer.

Fig. 9 shows reasoning traces of the baseline model (Qwen2.5-VL-3B) and models trained on SpatialBlock-15k with and without visual cues for a question from the MindCube dataset. The task requires aggregating visual information from three images captured from different viewpoints. In addition, the model should infer the spatial relationship between two objects from the perspective presented in the second image (image 2).

As shown in Fig. 9(a), the baseline model fails to exhibit multi-image understanding and merely describes image 2. The model trained without visual cues (Fig. 9(b)) attempts to reason over multiple images, but fails to identify the common object across views, leading to an incorrect conclusion. In contrast, SpatialBlock-3B-reason (Fig. 9(c)) identifies the “yellow container” as an anchor object and connects the given images through this shared reference, ultimately producing the correct answer.

These results demonstrate that training on our dataset enhances the model’s ability to perform complex spatial reasoning. Furthermore, incorporating visual cues enables the model

to effectively leverage anchor objects to integrate information across multiple viewpoints. Additional qualitative results are illustrated in Fig. 10.

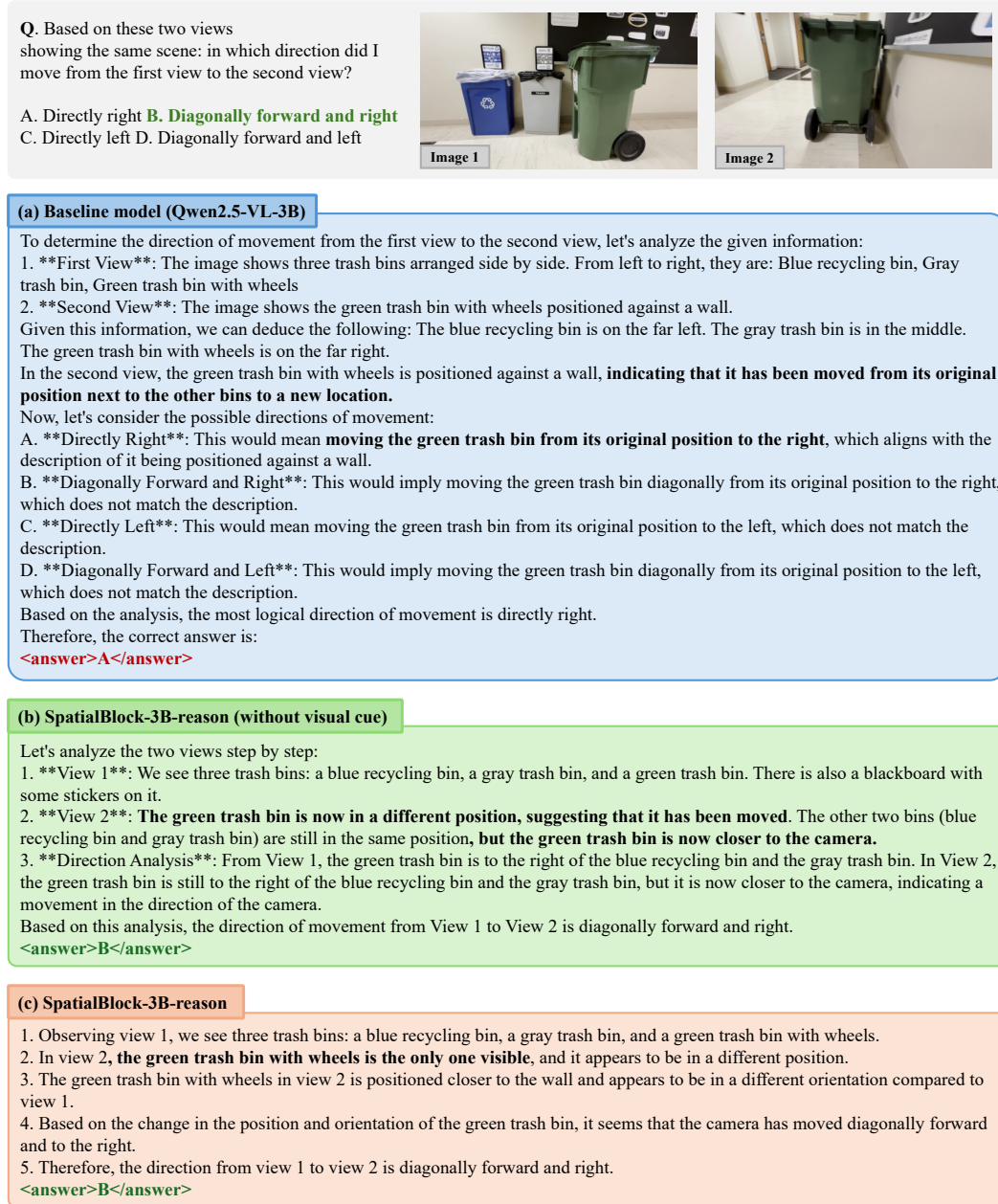


Figure 10: **Additional Qualitative Result on Reasoning Traces.** The figure shows additional qualitative results on reasoning traces of the baseline model, our model trained without visual cues, and our final model, SpatialBlock-3B-reason.